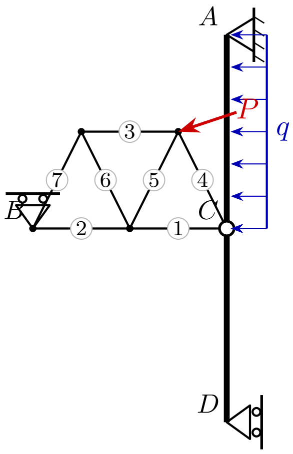
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k, reactions a, interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

2. Support reactions and hinge forces

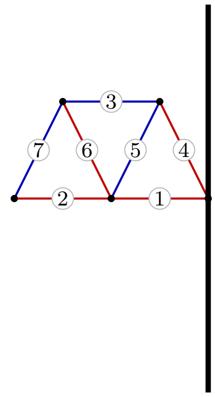
$A:\ R_x = 14.25,\ R_y = -3.75 \quad D:\ R_x = 9.75,\ R_y = 0 \quad B:\ R_x = 0,\ R_y = 8.75$

$C:\ H_x = -15,\ H_y = 3.75$

3. Truss member forces

Member	Force S [kN]	Type
1	13.12	tension
2	4.38	tension
3	-8.75	compression
4	4.19	tension
5	-9.78	compression
6	9.78	tension
7	-9.78	compression

red: tension (+) blue: compression (−)



4. Section-force diagrams of the beams